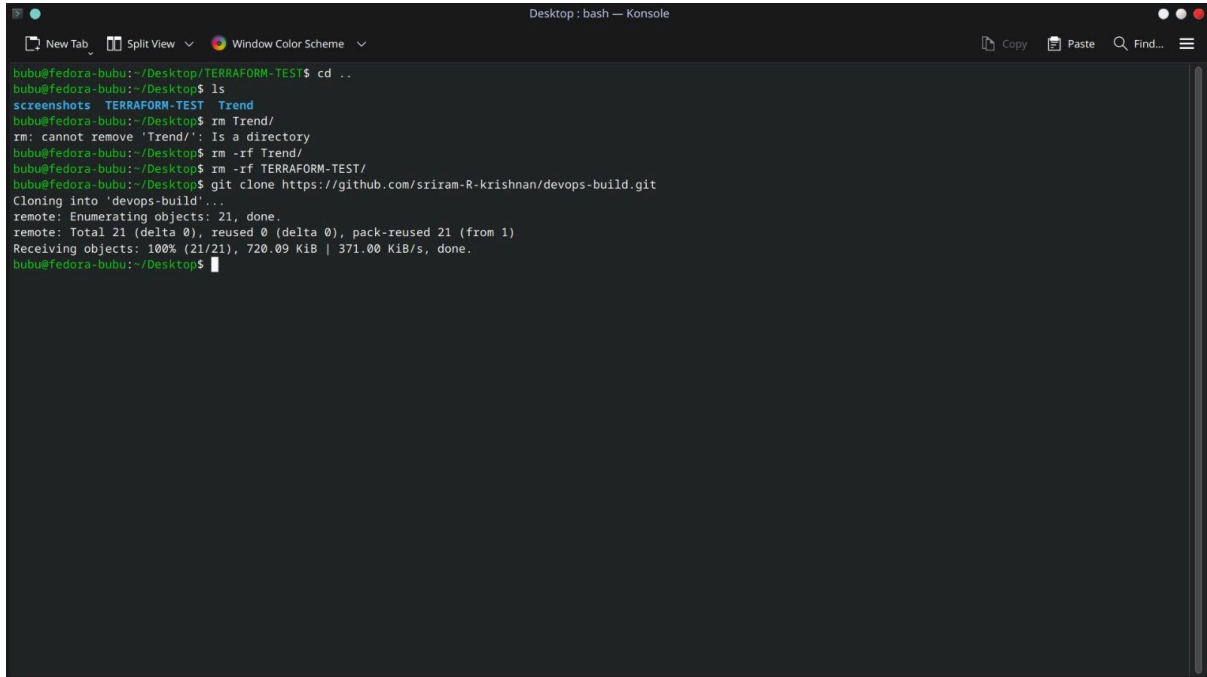


Reactjs E-commerce Application Documentation

Step 1 : Cloning the repository

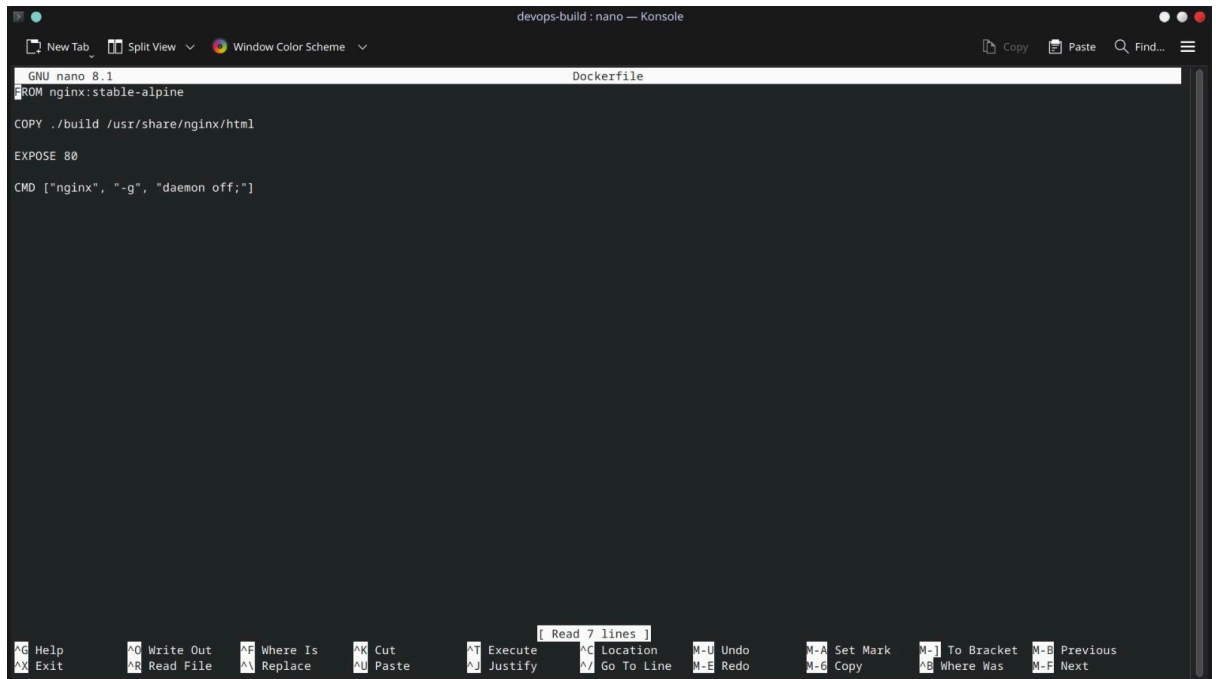
- Git clone <https://github.com/sriram-R-krishnan/devops-build>



```
Desktop: bash — Konsole
New Tab Split View Window Color Scheme Copy Paste Find...
bubu@fedora-bubu:~/Desktop/TERRAFORM-TEST$ cd ..
bubu@fedora-bubu:~/Desktop$ ls
screenshots  TERRAFORM-TEST  Trend
bubu@fedora-bubu:~/Desktop$ rm Trend/
rm: cannot remove 'Trend/': Is a directory
bubu@fedora-bubu:~/Desktop$ rm -rf Trend/
bubu@fedora-bubu:~/Desktop$ rm -rf TERRAFORM-TEST/
bubu@fedora-bubu:~/Desktop$ git clone https://github.com/sriram-R-krishnan/devops-build.git
Cloning into 'devops-build'...
remote: Enumerating objects: 21, done.
remote: Total 21 (delta 0), reused 0 (delta 0), pack-reused 21 (from 1)
Receiving objects: 100% (21/21), 720.09 KiB | 371.00 KiB/s, done.
bubu@fedora-bubu:~/Desktop$
```

Step 2 : Dockerize the application using dockerfile

```
FROM nginx:stable-alpine
COPY ./build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The terminal is running the GNU nano 8.1 editor, editing a file named "Dockerfile". The content of the Dockerfile is as follows:

```
FROM nginx:stable-alpine
COPY ./build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

The terminal interface includes a top bar with "New Tab", "Split View", and "Window Color Scheme" options. On the right, there are icons for "Copy", "Paste", "Find...", and a menu icon. The bottom status bar displays various keyboard shortcuts for nano, such as "H-G Help", "W-O Write Out", "W-I Where Is", "C Cut", "E Execute", "L Location", "U Undo", "S-M Set Mark", "T-B To Bracket", "P Previous", "X Exit", "R Read File", "A Replace", "D Paste", "J Justify", "G Go To Line", "E Redo", "C Copy", "W Where Was", and "N Next". A small status indicator in the center of the bottom bar shows "Read 7 lines".

Step 3 : docker-compose file

version: '3.8'

services:

web:

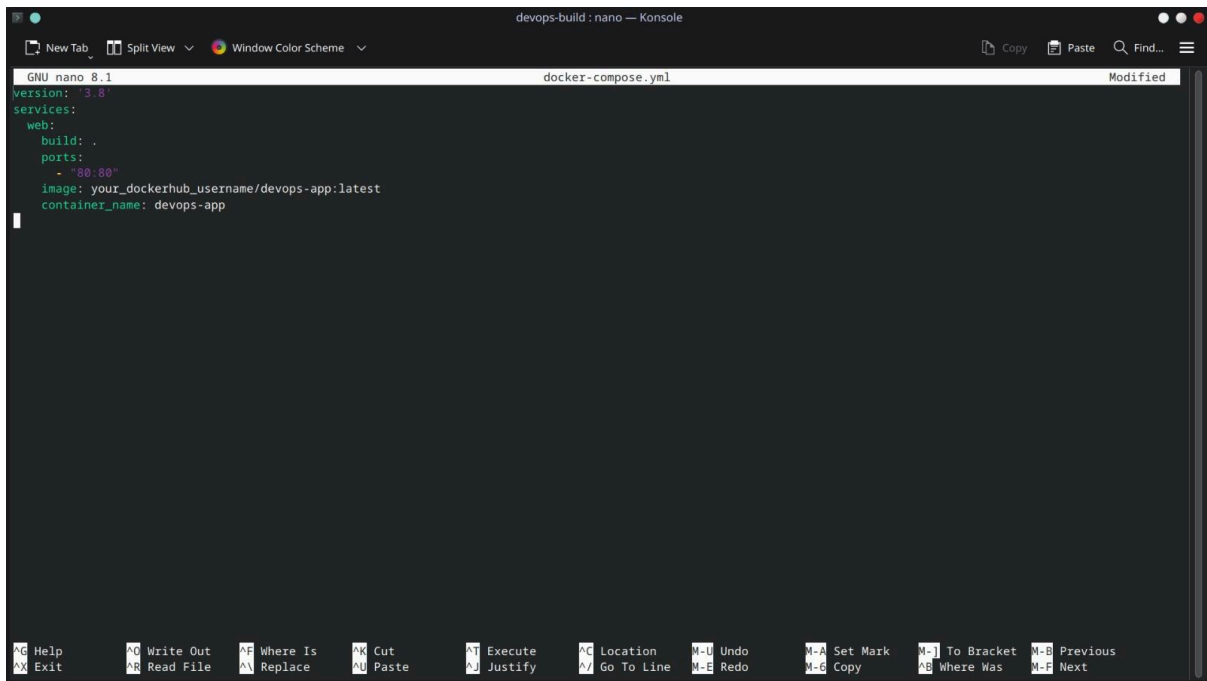
build: .

ports:

- "80:80"

image: your_dockerhub_username/devops-app:latest

container_name: devops-app



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The window displays the content of a file named "docker-compose.yml". The text in the file is as follows:

```
GNU nano 8.1 docker-compose.yml
version: '3.8'
services:
  web:
    build: .
    ports:
      - "80:80"
    image: your_dockerhub_username/devops-app:latest
    container_name: devops-app
```

The terminal window includes a menu bar at the top with options like "New Tab", "Split View", and "Window Color Scheme". At the bottom, there is a detailed help menu listing various keyboard shortcuts and their functions, such as "Help", "Exit", "Write Out", "Read File", "Where Is", "Replace", "Cut", "Paste", "Execute", "Justify", "Location", "Go To Line", "Undo", "Redo", "Set Mark", "Copy", "To Bracket", "Where Was", "Previous", and "Next".

Step 4 : Bash scripting

- Create build.sh and deploy.sh and make it executable

build.sh

```
#!/bin/bash
```

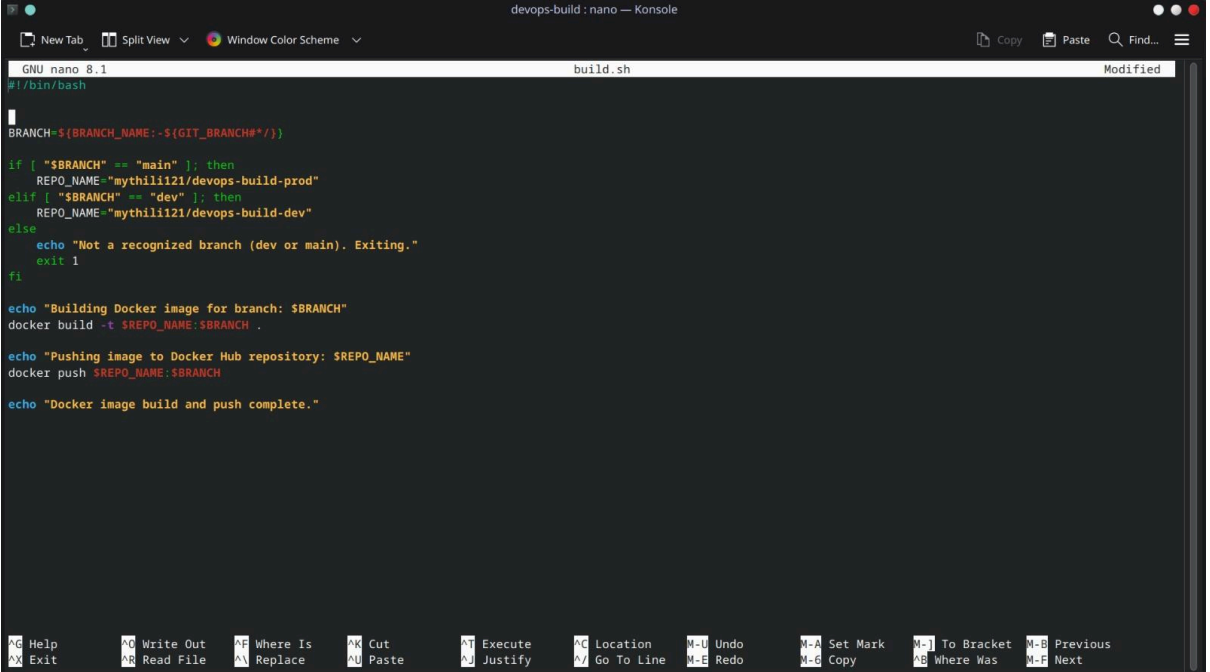
```
# Prefer BRANCH_NAME if set (Multibranch), otherwise fallback to GIT_BRANCH
BRANCH=${BRANCH_NAME:-${GIT_BRANCH#*/}}
```

```
if [ "$BRANCH" == "main" ]; then
    REPO_NAME="mythili121/devops-build-prod"
elif [ "$BRANCH" == "dev" ]; then
    REPO_NAME="mythili121/devops-build-dev"
else
    echo "Not a recognized branch (dev or main). Exiting."
    exit 1
fi
```

```
echo "Building Docker image for branch: $BRANCH"
docker build -t $REPO_NAME:$BRANCH .
```

```
echo "Pushing image to Docker Hub repository: $REPO_NAME"
docker push $REPO_NAME:$BRANCH
```

```
echo "Docker image build and push complete."
```



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The window displays the content of the "build.sh" script being edited in the nano text editor. The script includes logic to determine the repository name based on the current branch, build the Docker image, and push it to Docker Hub. The terminal interface includes standard nano editor controls at the top and bottom, and a status bar at the bottom showing various keyboard shortcuts.

```
GNU nano 8.1 build.sh
#!/bin/bash

BRANCH=${BRANCH_NAME:-${GIT_BRANCH#*/}}

if [ "$BRANCH" == "main" ]; then
    REPO_NAME="mythili121/devops-build-prod"
elif [ "$BRANCH" == "dev" ]; then
    REPO_NAME="mythili121/devops-build-dev"
else
    echo "Not a recognized branch (dev or main). Exiting."
    exit 1
fi

echo "Building Docker image for branch: $BRANCH"
docker build -t $REPO_NAME:$BRANCH .

echo "Pushing image to Docker Hub repository: $REPO_NAME"
docker push $REPO_NAME:$BRANCH

echo "Docker image build and push complete."
```

deploy.sh

```
#!/bin/bash
```

```
REPO_TYPE=$1 # e.g., "prod" or "dev"
```

```
TAG=$2      # e.g., "main" or "dev"
```

```
IMAGE_NAME="mythili121/devops-build-${REPO_TYPE}:${TAG}"
```

```
echo "Pulling the latest Docker image: $IMAGE_NAME"
```

```
docker pull $IMAGE_NAME
```

```
echo "Stopping and removing existing container (if any)..."
```

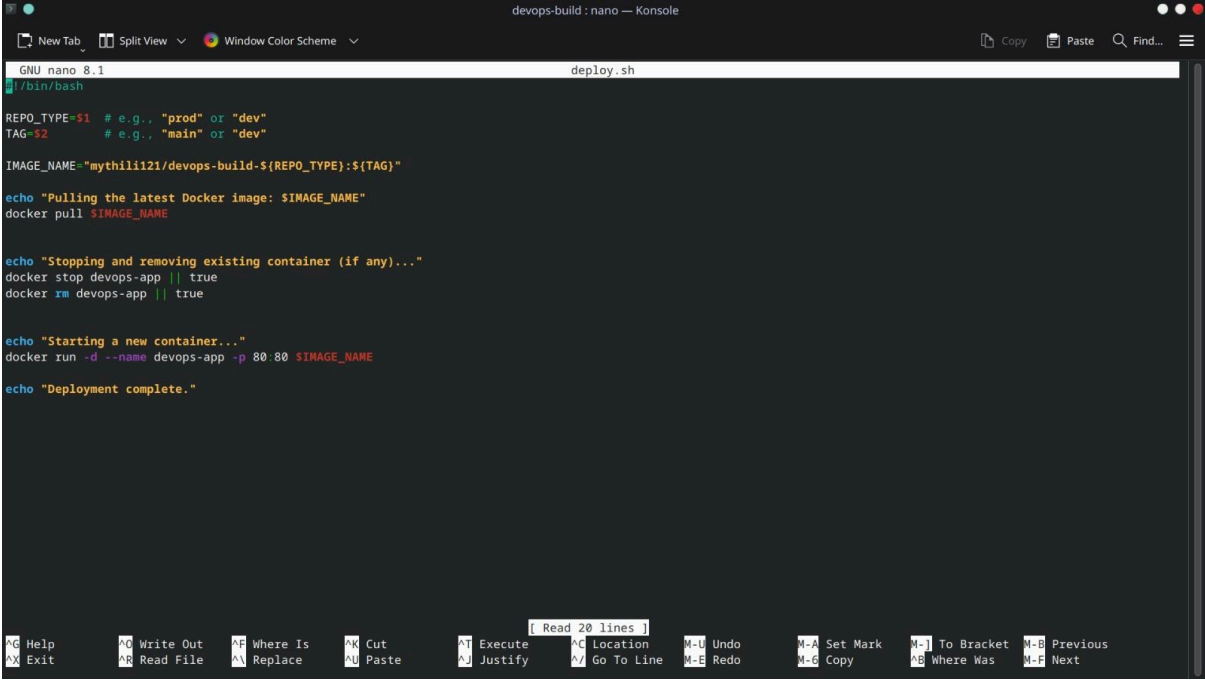
```
docker stop devops-app || true
```

```
docker rm devops-app || true
```

```
echo "Starting a new container..."
```

```
docker run -d --name devops-app -p 80:80 $IMAGE_NAME
```

```
echo "Deployment complete."
```



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The terminal displays the contents of the "deploy.sh" script, which is being edited in nano 8.1. The script includes comments for repository type and tag, sets the image name, pulls the latest Docker image, stops and removes any existing container, and finally runs a new container named "devops-app" on port 80. The terminal output shows the script being executed line by line, with the final output being "Deployment complete." The bottom of the terminal shows a list of nano editor shortcuts.

```
GNU nano 8.1 deploy.sh
#!/bin/bash

REPO_TYPE=$1 # e.g., "prod" or "dev"
TAG=$2      # e.g., "main" or "dev"

IMAGE_NAME="mythili121/devops-build-${REPO_TYPE}:${TAG}"

echo "Pulling the latest Docker image: $IMAGE_NAME"
docker pull $IMAGE_NAME

echo "Stopping and removing existing container (if any)..."
docker stop devops-app || true
docker rm devops-app || true

echo "Starting a new container..."
docker run -d --name devops-app -p 80:80 $IMAGE_NAME

echo "Deployment complete."
```

Read 20 lines

Alt-H Help	Alt-O Write Out	Alt-F Where Is	Alt-K Cut	Alt-T Execute	Alt-L Location	Alt-U Undo	Alt-A Set Mark	Alt-J To Bracket	Alt-B Previous
Alt-X Exit	Alt-R Read File	Alt-V Replace	Alt-N Paste	Alt-J Justify	Alt-G Go To Line	Alt-E Redo	Alt-C Copy	Alt-W Where Was	Alt-F Next

Step 5 : Version Control

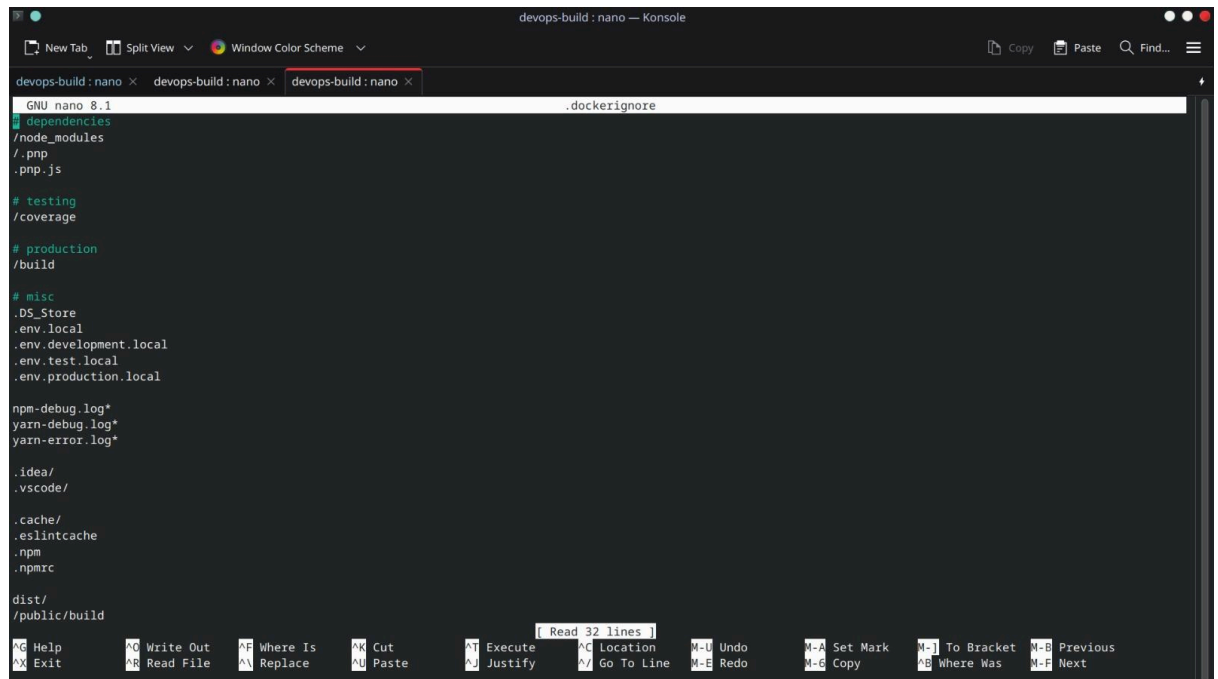
- Work on features locally, commit and push to the dev branch.
- dev branch: continuous integration builds and pushes to Docker Hub dev repo.

Dockerignore:

```
.git  
.gitignore  
.dockerignore  
Node_modules  
Npm-debug.log
```

README.md

```
**/*.  
.vscode  
.idea
```



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The editor displays the contents of a file named ".dockerignore". The file lists various directories and files to be ignored by Docker, including ".git", ".gitignore", ".dockerignore", "Node_modules", "Npm-debug.log", ".vscode", ".idea", ".cache", ".eslintrc", ".npm", ".npmrc", "dist", and "public/build". The terminal also shows the nano editor's command palette at the bottom with various shortcuts like "Help", "Write Out", "Where Is", "Cut", "Execute", "Location", "Undo", "Set Mark", "To Bracket", "Previous", "Exit", "Read File", "Replace", "Paste", "Justify", "Go To Line", "Redo", "Copy", "Where Was", and "Next".

```
GNU nano 8.1 .dockerignore  
# dependencies  
/node_modules  
/.pnp  
.pnp.js  
  
# testing  
/coverage  
  
# production  
/build  
  
# misc  
.DS_Store  
.env.local  
.env.development.local  
.env.test.local  
.env.production.local  
  
npm-debug.log*  
yarn-debug.log*  
yarn-error.log*  
  
.idea/  
.vscode/  
  
.cache/  
.eslintrc  
.npm  
.npmrc  
  
dist/  
public/build
```

gitignore:

```
# dependencies  
/node_modules  
/.pnp  
.pnp.js  
  
# testing  
/coverage  
  
# production
```

/build

misc

.DS_Store

.env.local

.env.development.local

.env.test.local

.env.production.local

npm-debug.log*

yarn-debug.log*

yarn-error.log*

.idea/

.vscode/

.cache/

.eslintcache

.npm

.npmrc

dist/

/public/build



The screenshot shows a terminal window titled "devops-build : nano — Konsole". The window displays the contents of a file named ".dockerignore" using the nano text editor. The file contains the following lines:

```
git
.gitignore
.dockerignore
node_modules
npm-debug.log
README.md
**/.
.vscode
.idea
```

The terminal window includes a top menu bar with options like "New Tab", "Split View", and "Window Color Scheme". A bottom status bar shows various keyboard shortcuts for nano, such as "H Help", "W Write Out", "F Where Is", "K Cut", "T Execute", "C Location", "U Undo", "A Set Mark", "I To Bracket", "B Previous", "X Exit", "R Read File", "N Replace", "P Paste", "J Justify", "G Go To Line", "E Redo", "G Copy", "B Where Was", and "F Next". A status indicator in the center of the bottom bar reads "[Read 9 lines]".

Git commands

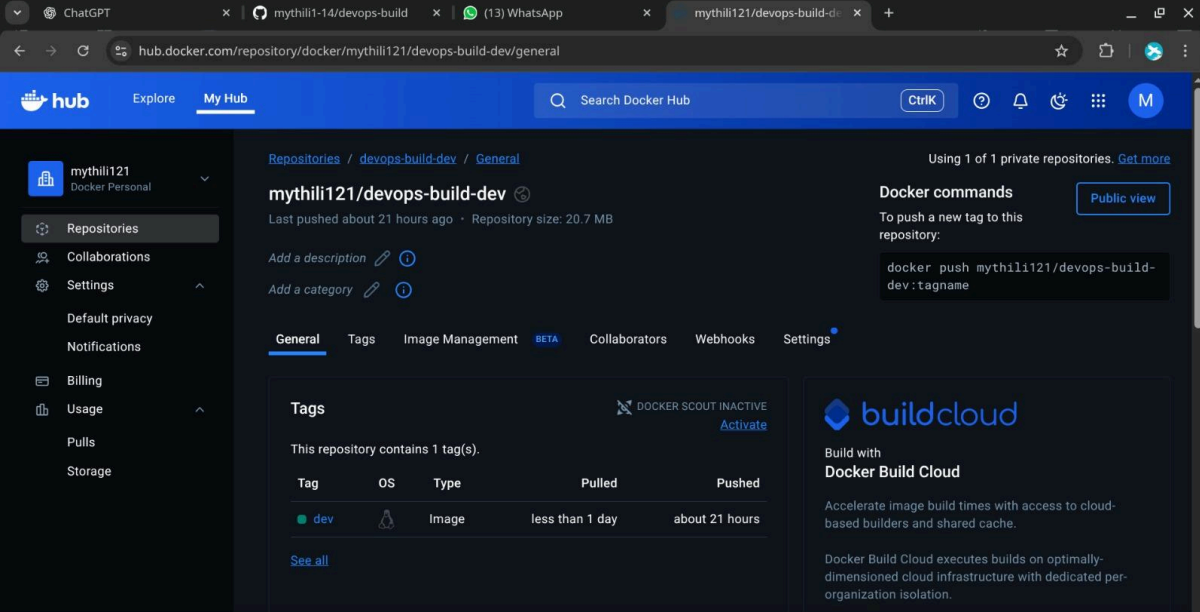
- `git add .`
- `git commit -m "feat: add X"`
- `git push origin dev`
- # When merged to master:
- `git checkout master`
- `git pull origin master`

```
devops-build : bash — Konsole
New Tab Split View Window Color Scheme
Copy Paste Find...
bubu@fedora-bubu:~/Desktop/devops-build$ git remote set-url origin https://github.com/mythili1-14/devops-build.git
bubu@fedora-bubu:~/Desktop/devops-build$ git branch
* main
bubu@fedora-bubu:~/Desktop/devops-build$ git branch -c dev
bubu@fedora-bubu:~/Desktop/devops-build$ git branch
dev
* main
bubu@fedora-bubu:~/Desktop/devops-build$ git checkout dev
Switched to branch 'dev'
Your branch is up to date with 'origin/main'.
bubu@fedora-bubu:~/Desktop/devops-build$ git branch
* dev
main
bubu@fedora-bubu:~/Desktop/devops-build$
```

```
devops-build : bash — Konsole
New Tab Split View Window Color Scheme
Copy Paste Find...
bubu@fedora-bubu:~/Desktop/devops-build$ git push -u origin dev
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 12 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (21/21), 720.09 KiB | 144.02 MiB/s, done.
Total 21 (delta 0), reused 21 (delta 0), pack-reused 0 (from 0)
To https://github.com/mythili1-14/devops-build.git
 * [new branch]      dev -> dev
branch 'dev' set up to track 'origin/dev'.
bubu@fedora-bubu:~/Desktop/devops-build$
```


Step 6 : Docker hub:

- Create two repositories:
 - devops-build-dev — public
 - devops-build-prod — private
- Use docker tag and docker push to upload images

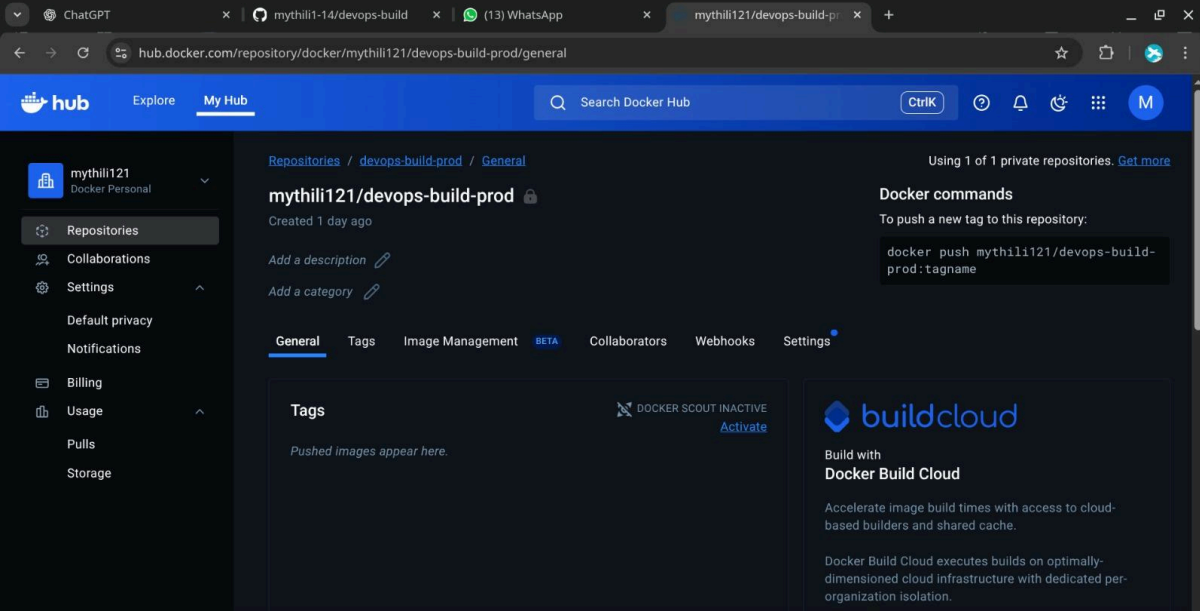


The screenshot shows the Docker Hub interface for the repository `mythili121/devops-build-dev`. The page is titled "mythili121/devops-build-dev" and indicates it was last pushed about 21 hours ago with a repository size of 20.7 MB. The "General" tab is selected, showing a table of tags. The table has columns for Tag, OS, Type, Pulled, and Pushed. One tag, `dev`, is listed with OS `linux` and Type `Image`. The "Pushed" column shows "about 21 hours". A "Docker commands" section on the right provides the command `docker push mythili121/devops-build-dev:tagname`. A "buildcloud" advertisement is also visible.

Tag	OS	Type	Pulled	Pushed
dev	linux	Image	less than 1 day	about 21 hours

By clicking "Accept All Cookies", you agree to the storing of cookies on your device to enhance site navigation, analyze site usage, and assist in our marketing efforts.

[Cookies Settings](#) [Reject All](#) [Accept All Cookies](#)



The screenshot shows the Docker Hub interface for the repository `mythili121/devops-build-prod`. The page is titled "mythili121/devops-build-prod" and indicates it was created 1 day ago. The "General" tab is selected, showing a message "Pushed images appear here." The "Docker commands" section on the right provides the command `docker push mythili121/devops-build-prod:tagname`. A "buildcloud" advertisement is also visible.

Pushed images appear here.

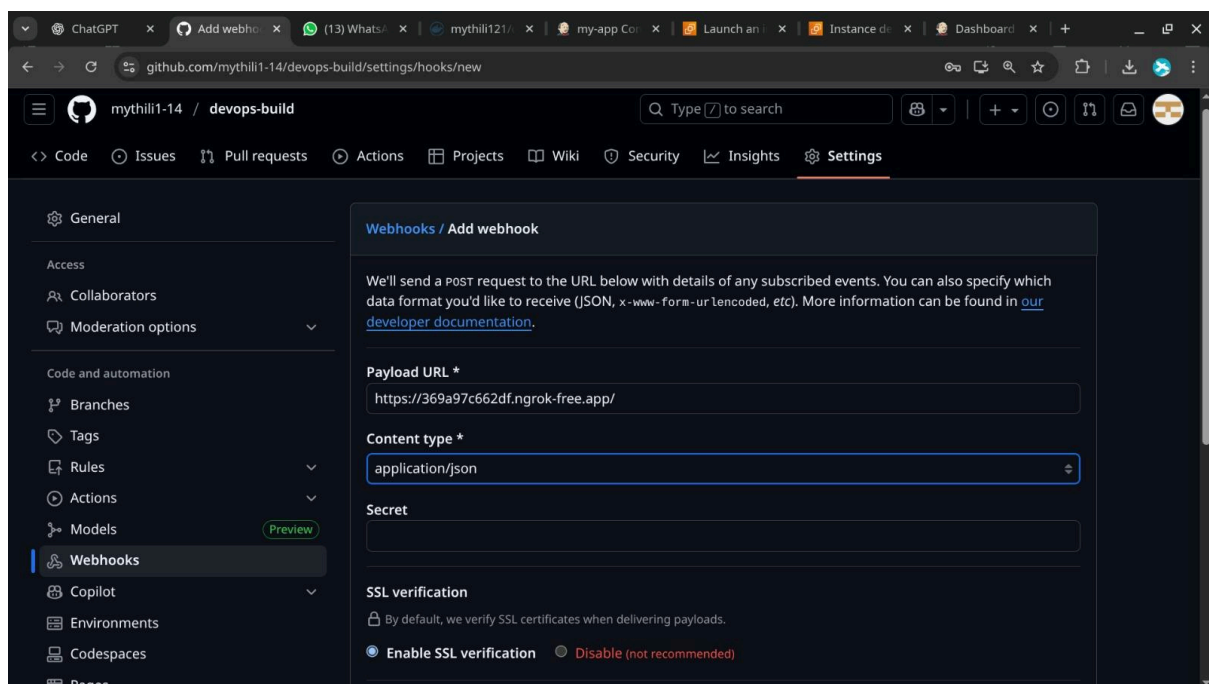
By clicking "Accept All Cookies", you agree to the storing of cookies on your device to enhance site navigation, analyze site usage, and assist in our marketing efforts.

[Cookies Settings](#) [Reject All](#) [Accept All Cookies](#)

```
devops-build : bash — Konsole
New Tab Split View Window Color Scheme Copy Paste Find...
nothing added to commit but untracked files present (use "git add" to track)
bubu@fedora-bubu:~/Desktop/devops-build$ git add .
bubu@fedora-bubu:~/Desktop/devops-build$ git commit -m "initial commit" .
[dev c061fc1] initial commit
6 files changed, 120 insertions(+)
create mode 100644 .dockerignore
create mode 100644 .gitignore.save
create mode 100644 Dockerfile
create mode 100644 build.sh.save
create mode 100644 deploy.sh
create mode 100644 docker-compose.yml
bubu@fedora-bubu:~/Desktop/devops-build$ git push origin dev
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.37 KiB | 1.37 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mythili1-14/devops-build.git
ef44730..c061fc1 dev -> dev
bubu@fedora-bubu:~/Desktop/devops-build$
```

Step 7 : Jenkins

- Install Jenkins (on a server or VPS)
- Install Java 11+, install Jenkins LTS.
- Install plugins: GitHub, Docker, Pipeline, Credentials, GitHub Branch Source.
- Create credentials
 - Docker Hub credentials (username/password)
 - GitHub credentials/token (if private repo)
 - SSH key credential for EC2 (if using ssh deploy)
- Configure webhooks
 - In GitHub repo settings → Webhooks: add Jenkins webhook URL so pushing to dev or master triggers build automatically.



Jenkinsfile:

```
pipeline {
  agent any
  environment {
    DOCKERHUB_USERNAME = 'mythili121'
    DOCKERHUB_CREDENTIAL_ID = 'dockerhub-cred'
    AWS_SSH_CREDENTIAL_ID = 'aws-ec2-key'
    EC2_IP = '54.172.118.89'
  }

  stages {
    stage('Build and Push') {
      steps {
        withCredentials([usernamePassword(credentialsId:
DOCKERHUB_CREDENTIAL_ID, usernameVariable: 'DOCKER_USERNAME',
passwordVariable: 'DOCKER_PASSWORD')]) {
          sh "echo \${DOCKER_PASSWORD} | docker login -u \${DOCKER_USERNAME}
--password-stdin"
        }
        sh "bash ./build.sh"
      }
    }

    stage('Deploy to Server') {
      steps {
        sshagent(credentials: [AWS_SSH_CREDENTIAL_ID]) {
          sh "scp -o StrictHostKeyChecking=no deploy.sh
ubuntu@${EC2_IP}:/tmp/deploy.sh"
          sh "ssh -o StrictHostKeyChecking=no ubuntu@${EC2_IP} bash
/tmp/deploy.sh ${env.BRANCH_NAME == 'main' ? 'prod main' : 'dev dev'}"
        }
      }
    }
  }
}
```

ChatGPT x devops-bui x (14) Whats x mythili121 x Launch an x Instance de x SecurityGre x Download x

Not secure 98.84.168.134:8080/manage/pluginManager/updates/

☆ | | | |

Jenkins

Manage Jenkins

Plugins

Search

Settings

Help

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

commons-lang3 v3.x Jenkins API Success

Icons API Success

Folders Success

OWASP Markup Formatter Success

ASM API Success

JSON Path API Success

Struts Success

Pipeline: Step API Success

commons-text API Success

Token Macro Success

Build Timeout Success

bouncycastle API Success

Credentials Success

Plain Credentials Success

Variant Success

SSH Credentials Success

Credentials Binding Success

ChatGPT x mythili1-14/devops-build x (13) WhatsApp x mythili121/devops-build x Download progress - Plug x

localhost:8080/manage/pluginManager/updates/

☆ | | | |

Jenkins

Manage Jenkins

Plugins

Search

Settings

Menu

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

SSH Agent Success

Loading plugin extensions Success

Go back to the top page

(you can start using the installed plugins right away)

Restart Jenkins when installation is complete and no jobs are running

ChatGPT x mythili1- x devops-l x (13) Wha x mythili12 x Personal x System x Launch x Instance x +

localhost:8080/manage/credentials/store/system/domain/

localhost:8080/manage/credentials/store/system/domain/

Jenkins

Manage Jenkins

Credentials

System

Global credentials (unrestricted)

?

?

?

Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
 aws-ec2-key	ubuntu	SSH Username with private key	
 dockerhub-cred	mythili121/*****	Username with password	

Icon: S M L

REST API Jenkins 2.516.1

ChatGPT x mythili1-14/de x (13) WhatsApp x mythili121/dev x Launch an inst x Instance detai x my-app Config x +

369a97c662df.ngrok-free.app/view/all/job/my-app/configure

369a97c662df.ngrok-free.app/view/all/job/my-app/configure

Jenkins

All

my-app

Configuration

?

?

?

Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/mythili1-14/devops-build.git

Credentials ?

- none -

Save

Apply

ChatGPTmythili1-14/de(13) WhatsAppmythili121/devLaunch an instInstance detailmy-app Config

369a97c662df.ngrok-free.app/view/all/job/my-app/configure

JenkinsAllmy-appConfiguration

Configure

General

Triggers

Pipeline

Advanced

Branch Specifier (blank for 'any') ?

* /main

Add Branch

Repository browser ?

(Auto)

Additional Behaviours

Add

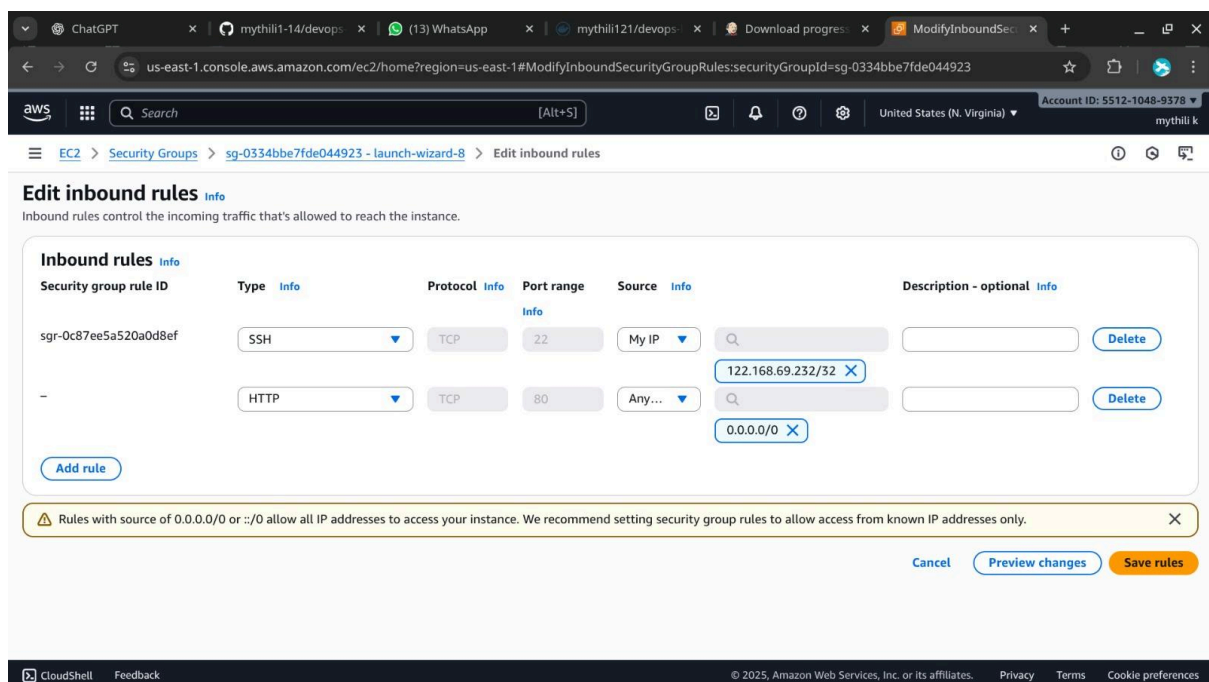
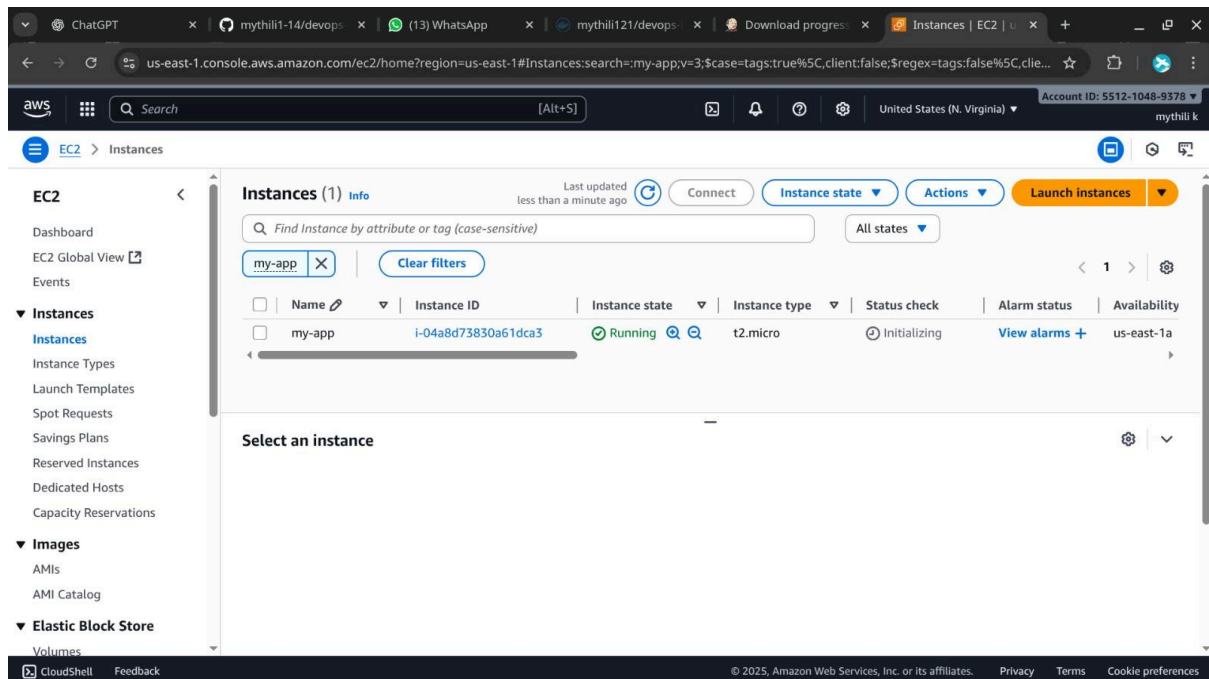
Script Path ?

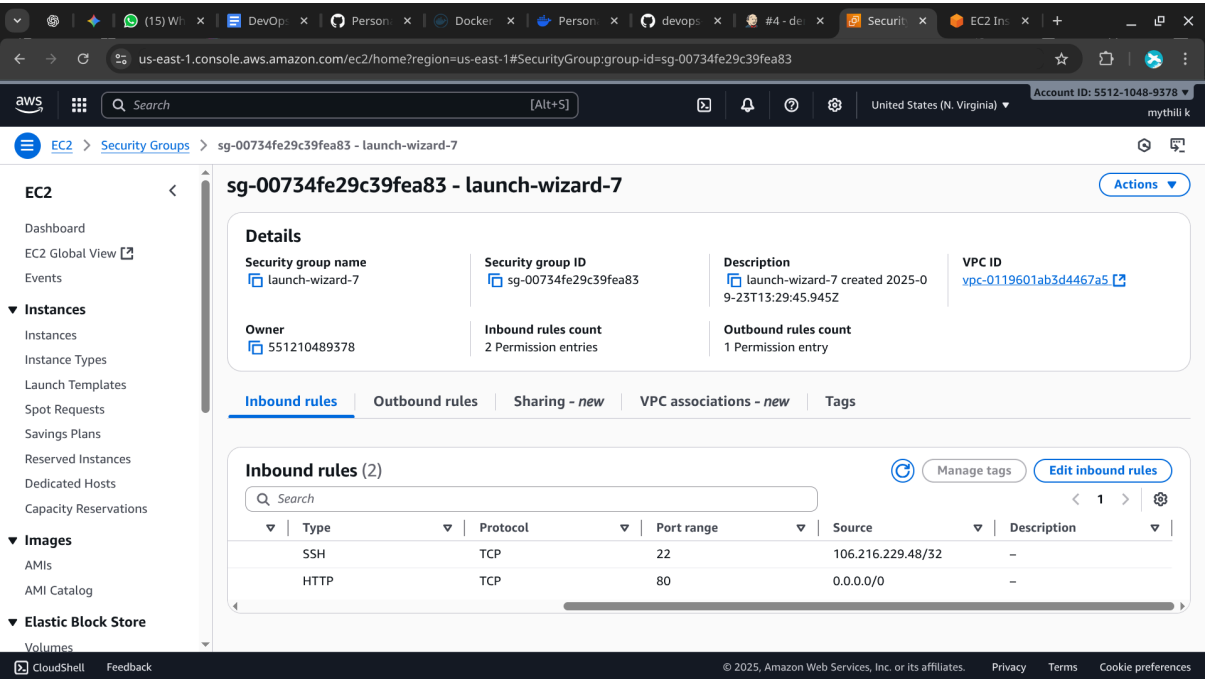
jenkinsfile

SaveApply

Step 8 : AWS

- Launch an t2.micro EC2 instance (Amazon Linux 2 or Ubuntu).
- Configure the Security Group (SG):
 - HTTP (TCP 80) — Source: 0.0.0.0/0 (or restrict to required IP ranges if desired).
 - SSH (TCP 22) — Source: your IP only for security.
- SSH to instance and install Docker & docker-compose.
- Pull image & run using docker-compose.yml on server and docker-compose up -d.





```
(ubuntu) 98.84.168.134 — Konsole
New Tab Split View Window Color Scheme Copy Paste Find...
bubu@fedora-bubu:~/Downloads$ ssh -i aws-ec2-key.pem ubuntu@98.84.168.134
The authenticity of host '98.84.168.134 (98.84.168.134)' can't be established.
ED25519 key fingerprint is SHA256:F1I0TeVIAwOvg7TvRQqxKS2d1HtFD0C7JmNlmKC+x4k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.84.168.134' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@                WARNING: UNPROTECTED PRIVATE KEY FILE!                @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'aws-ec2-key.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "aws-ec2-key.pem": bad permissions
ubuntu@98.84.168.134: Permission denied (publickey).
bubu@fedora-bubu:~/Downloads$ chmod 700 aws-ec2-key.pem
bubu@fedora-bubu:~/Downloads$ ssh -i aws-ec2-key.pem ubuntu@98.84.168.134
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

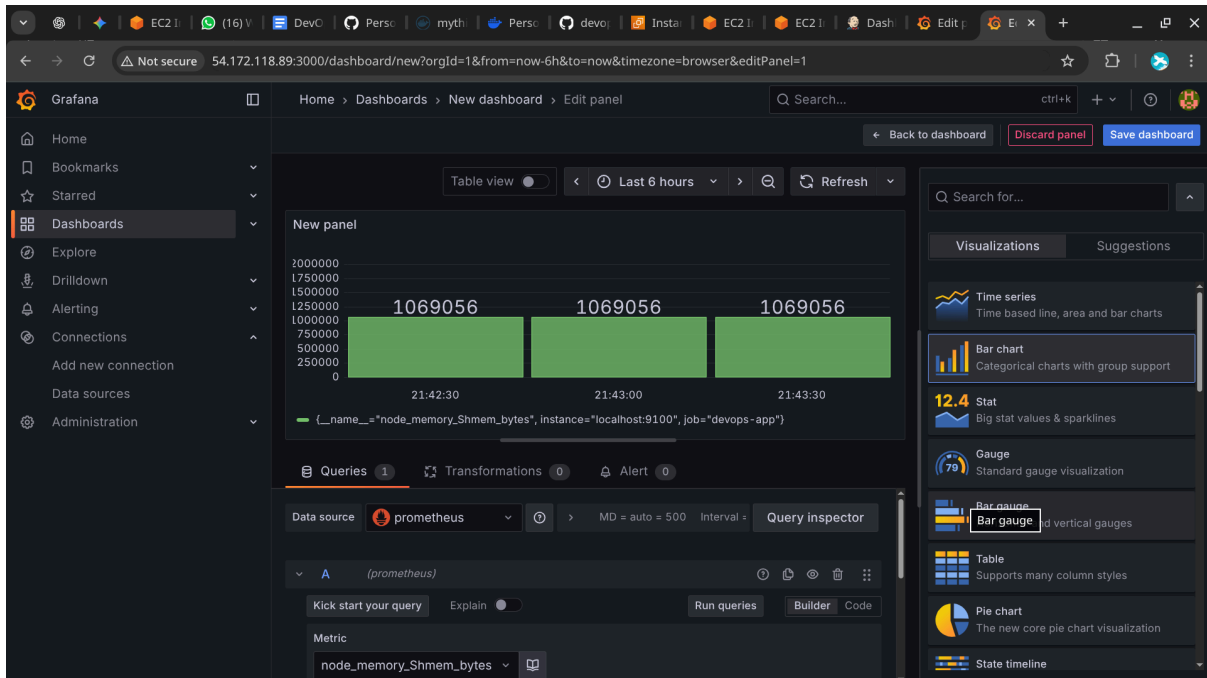
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

```
(ubuntu) 98.84.168.134 — Konsole
New Tab Split View Window Color Scheme Copy Paste Find...
devops-build: bash x (ubuntu) 98.84.168.134 x
root@ip-172-31-24-39:/home/ubuntu# apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 37 not upgraded.
Need to get 79.2 MB of archives.
After this operation, 300 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.2.5-0ubuntu1-24.04.1 [8043 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.7.27-0ubuntu1-24.04.1 [37.7 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dns-root-data all 2024071801-ubuntu0.24.04.1 [5918 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dnsmasq-base amd64 2.90-2ubuntu0.1 [376 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 docker.io amd64 27.5.1-0ubuntu3-24.04.2 [33.0 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 ubuntu-fan all 0.12.16+24.04.1 [34.2 kB]
Fetched 79.2 MB in 1s (85.4 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-1ubuntu2_amd64.deb ...
Unpacking bridge-utils (1.7.1-1ubuntu2) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.2.5-0ubuntu1-24.04.1_amd64.deb ...
Unpacking runc (1.2.5-0ubuntu1-24.04.1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_1.7.27-0ubuntu1-24.04.1_amd64.deb ...
```

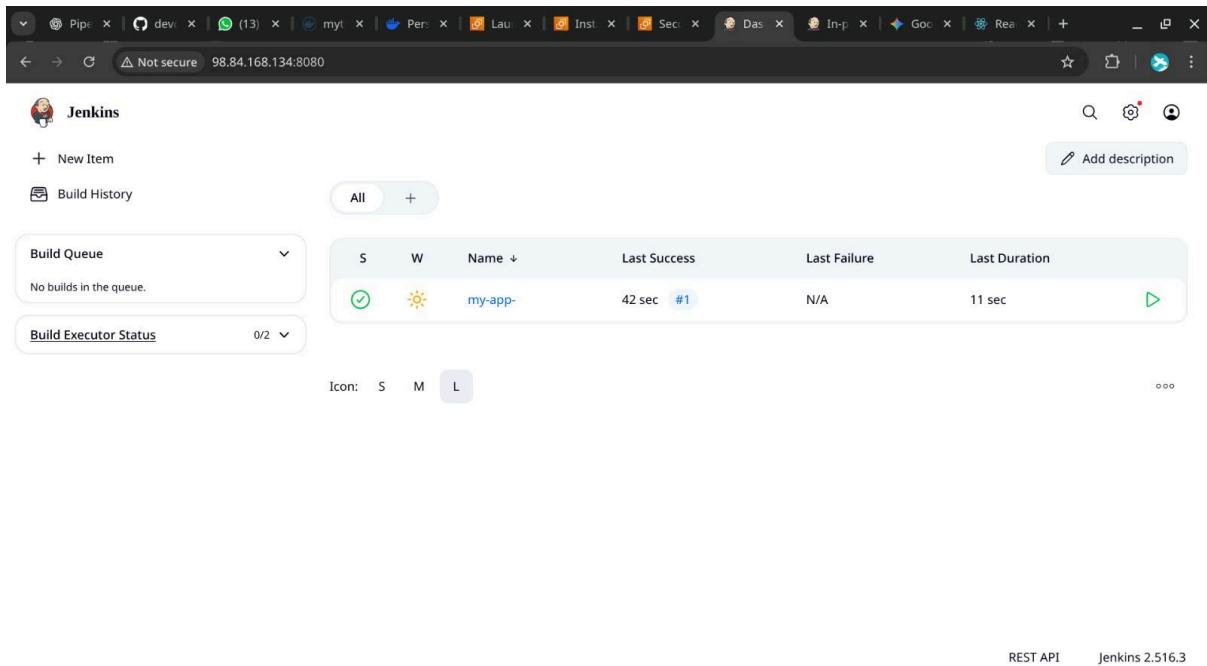
```
devops-build : bash — Konsole
New Tab Split View Window Color Scheme
Copy Paste Find...
build build.sh deploy.sh docker-compose.yml Dockerfile Jenkinsfile
bubu@fedora-bubu:~/Desktop/devops-build$ git pull origin dev
From https://github.com/mythili1-14/devops-build
* branch          dev          -> FETCH_HEAD
Already up to date.
bubu@fedora-bubu:~/Desktop/devops-build$ git add .
bubu@fedora-bubu:~/Desktop/devops-build$ git commit -m "added Jenkinsfile" Jenkinsfile
[dev 3768664] added Jenkinsfile
1 file changed, 30 insertions(+)
create mode 100644 Jenkinsfile
bubu@fedora-bubu:~/Desktop/devops-build$ git push origin dev
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 730 bytes | 730.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mythili1-14/devops-build.git
45f2575..3768664 dev -> dev
bubu@fedora-bubu:~/Desktop/devops-build$
```

Step 9: Monitoring

- Install Prometheus to scrape Node Exporter or app metrics, and Grafana for dashboards.
- Add alertmanager for email/Slack alerts when app is unhealthy



Step 10 : Output

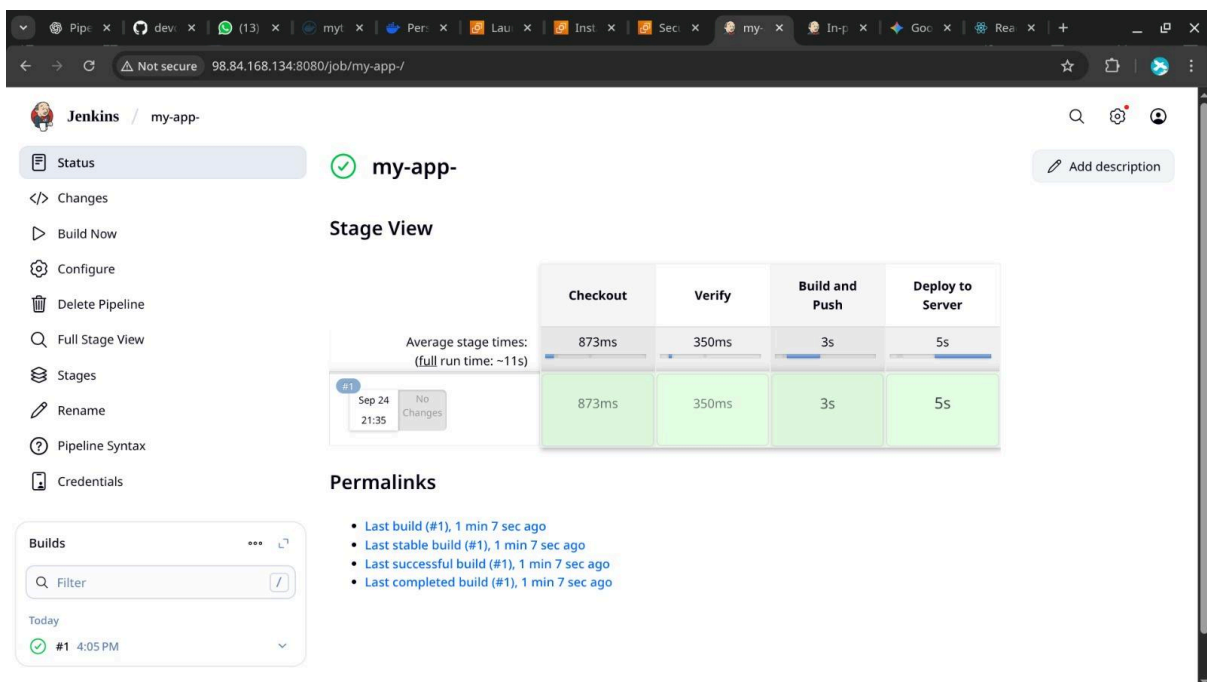


A screenshot of the Jenkins web interface. The top navigation bar includes a search icon, a settings gear, and a user profile icon. Below the navigation bar, there's a 'New Item' button and an 'Add description' button. The main content area shows a table of builds. The table has columns for 'S' (Status), 'W' (Webhook), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. A single build is listed with a green status icon, a sun icon, the name 'my-app-', a last success time of '42 sec', a last failure of 'N/A', and a last duration of '11 sec'. To the left of the table, there are two dropdown menus: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). At the bottom right, there's a footer with 'REST API' and 'Jenkins 2.516.3'.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☀	my-app-	42 sec #1	N/A	11 sec

Icon: S M L

REST API Jenkins 2.516.3



A screenshot of the Jenkins web interface showing the 'my-app-' pipeline view. The left sidebar contains a list of actions: 'Status' (selected), 'Changes', 'Build Now', 'Configure', 'Delete Pipeline', 'Full Stage View', 'Stages', 'Rename', 'Pipeline Syntax', and 'Credentials'. The main content area shows the 'my-app-' pipeline status as 'Success'. Below this, there's a 'Stage View' section with a table showing stage times. The table has columns for 'Checkout', 'Verify', 'Build and Push', and 'Deploy to Server'. The average stage times are: Checkout (873ms), Verify (350ms), Build and Push (3s), and Deploy to Server (5s). The full run time is ~11s. Below the stage view, there's a 'Permalinks' section with a list of links: 'Last build (#1), 1 min 7 sec ago', 'Last stable build (#1), 1 min 7 sec ago', 'Last successful build (#1), 1 min 7 sec ago', and 'Last completed build (#1), 1 min 7 sec ago'. At the bottom left, there's a 'Builds' section with a filter input and a list of builds. The first build is '#1 4:05 PM' with a green status icon.

Stage	Checkout	Verify	Build and Push	Deploy to Server
Average stage times: (full run time: ~11s)	873ms	350ms	3s	5s
#1 Sep 24 21:35 No Changes	873ms	350ms	3s	5s

Permalinks

- Last build (#1), 1 min 7 sec ago
- Last stable build (#1), 1 min 7 sec ago
- Last successful build (#1), 1 min 7 sec ago
- Last completed build (#1), 1 min 7 sec ago

Builds

Filter

Today

#1 4:05 PM

Jenkins / my-app- / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

#1 (Sep 24, 2025, 4:05:56 PM)

Add description

Keep this build forever

Started by user Mythili

Started 1 min 24 sec ago
Took 11 sec

This run spent:

- 10 ms waiting;
- 11 sec build duration;
- 11 sec total from scheduled to completion.

git

Revision: 1015d2e896cddd1b9524026ea9360e6d0080caf5
Repository: <https://github.com/mythili1-14/devops-build.git>

- refs/remotes/origin/dev

No changes.

REST API Jenkins 2.516.3

Jenkins / demo-app / #4 / Pipeline Overview

#4

Manually run by mythili

Started 1 min 19 sec ago

Queued 4 ms

Took 10 sec

Rerun

Graph

Start Checkout SCM Build and Push Deploy to Server End

Search

Checkout SCM 0.39s

Build and Push 2.9s

Deploy to Server 2.54s

Deploy to Server

5.4s Started 1m 20s ago Jenkins

```
scp -o StrictHostKeyChecking=no deploy.sh ubuntu@54.172.118.89:/tmp/deploy.sh >
ssh -o StrictHostKeyChecking=no ubuntu@54.172.118.89 bash /tmp/deploy.sh dev dev
0 + ssh -o StrictHostKeyChecking=no ubuntu@54.172.118.89 bash /tmp/deploy.sh dev dev
1 Pulling the latest Docker image: mythili121/devops-build-dev:dev
2 dev: Pulling from mythili121/devops-build-dev
```


Jenkins / my-app- / #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console Output

Download Copy View as plain text

```
Started by user Mythili
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/my-app-
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/mythilil-14/devops-build.git
> git init /var/lib/jenkins/workspace/my-app- # timeout=10
Fetching upstream changes from https://github.com/mythilil-14/devops-build.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/mythilil-14/devops-build.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/mythilil-14/devops-build.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
```

OnlineShop

Home Login SignUp

OnePlus 9 5G  5.4 inch display 399 Add to Cart	Iphone 13 mini  5.4 inch display 399 Add to Cart	Samsung s21 ultra  5.4 inch display 399 Add to Cart	xiomi mi 11  5.4 inch display 399 Add to Cart
OnePlus 9 5G  5.4 inch display 399 Add to Cart	Iphone 13 mini  5.4 inch display 399 Add to Cart	Samsung s21 ultra  5.4 inch display 399 Add to Cart	xiomi mi 11  5.4 inch display 399 Add to Cart